

# Zero-Knowledge from Secure Multiparty Computation & Ligerio

*A Unified Project Report*

Covering:

[IKOS07] Ishai, Kushilevitz, Ostrovsky, Sahai. *Zero-Knowledge from Secure Multiparty Computation*. STOC 2007.

[AHIV22] Ames, Hazay, Ishai, Venkatasubramanian. *Ligerio: Lightweight Sublinear Arguments Without a Trusted Setup*. CCS 2017 / ePrint 2022/1608.

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# Contents

|                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| <b>Overview</b>                                                                | <b>2</b>  |
| <b>1 Introduction</b>                                                          | <b>2</b>  |
| 1.1 Background and Motivation . . . . .                                        | 2         |
| 1.2 Main Contributions of the Two Papers . . . . .                             | 2         |
| 1.3 Organisation of This Report . . . . .                                      | 3         |
| <b>2 Preliminaries and Definitions</b>                                         | <b>3</b>  |
| 2.1 Interactive Proof Systems and Zero-Knowledge . . . . .                     | 3         |
| 2.2 Commitment Schemes . . . . .                                               | 4         |
| 2.3 Secret Sharing . . . . .                                                   | 4         |
| 2.4 Secure Multiparty Computation . . . . .                                    | 5         |
| 2.5 Coin-Flipping Protocol . . . . .                                           | 5         |
| 2.6 Error-Correcting Codes . . . . .                                           | 6         |
| 2.7 Interactive PCP and IOP . . . . .                                          | 6         |
| <b>3 Zero-Knowledge from Secure MPC (IKOS07)</b>                               | <b>7</b>  |
| 3.1 Setting and High-Level Idea . . . . .                                      | 7         |
| 3.2 Overview and Goals . . . . .                                               | 7         |
| 3.3 The Functionality and MPC Protocol Setup . . . . .                         | 8         |
| 3.4 The Basic Protocol $\Pi_R$ . . . . .                                       | 8         |
| 3.5 Security Analysis of $\Pi_R$ . . . . .                                     | 8         |
| 3.5.1 Completeness . . . . .                                                   | 8         |
| 3.5.2 Soundness . . . . .                                                      | 9         |
| 3.5.3 Zero-Knowledge . . . . .                                                 | 9         |
| 3.6 Protocol $\Pi'_R$ : Handling Statistical Correctness . . . . .             | 10        |
| 3.7 Protocol $\Pi_{R,t}$ : Robust MPC and Single-Round Amplification . . . . . | 10        |
| 3.8 Efficiency Analysis . . . . .                                              | 11        |
| <b>4 Ligerio: Lightweight Sublinear ZK Arguments</b>                           | <b>11</b> |
| 4.1 Setting and High-Level Idea . . . . .                                      | 11        |
| 4.2 Overview and Goals . . . . .                                               | 11        |
| 4.3 The Damgård–Ishai MPC Protocol . . . . .                                   | 11        |
| 4.4 Interleaved Reed–Solomon Codes . . . . .                                   | 12        |
| 4.5 The Ligerio ZK Protocol . . . . .                                          | 12        |
| 4.6 Security of Ligerio . . . . .                                              | 13        |
| 4.7 Communication Complexity of Ligerio . . . . .                              | 13        |
| <b>5 Comparison: IKOS07 vs. Ligerio</b>                                        | <b>14</b> |
| <b>6 What the Papers Do Not Solve</b>                                          | <b>15</b> |
| 6.1 Limitations of IKOS07 . . . . .                                            | 15        |
| 6.2 Limitations of Ligerio . . . . .                                           | 15        |
| <b>7 Open Problems</b>                                                         | <b>16</b> |
| <b>8 Subsequent Work and Impact</b>                                            | <b>16</b> |

## Overview

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Zero-knowledge (ZK) proofs and secure multiparty computation (MPC) are two foundational primitives in modern cryptography. Although they address different problems, Ishai, Kushilevitz, Ostrovsky, and Sahai (STOC 2007) [IKOS07] established a deep conceptual link between them: *any* semi-honest MPC protocol can be converted in a black-box manner into a ZK proof system, giving the first *constant-rate* ZK proofs for all of NP under one-way functions. The central technique — “MPC-in-the-head” — has since become one of the most influential paradigms in applied cryptography.

Building directly on this foundation, Ames, Hazay, Ishai, and Venkatasubramanian (CCS 2017, extended ePrint 2022/1608) [AHIV22] introduced **Ligerio**: a concrete, practically efficient ZK argument whose communication complexity is  $\mathcal{O}(\sqrt{s})$ , where  $s$  is the circuit size, relying only on collision-resistant hash functions. Ligerio instantiates the MPC-in-the-head framework with an optimised Damgård–Ishai MPC protocol and “interleaved” Reed–Solomon codes, giving the first *sublinear* transparent ZK argument with no trusted setup.

This report surveys both papers in depth. We present the formal definitions, core protocols, key lemmas and their proofs, compare what each paper achieves and does not achieve, discuss open problems, and describe the rich line of subsequent work inspired by these results.

**Keywords:** zero-knowledge proofs, secure multiparty computation, MPC-in-the-head, interactive PCPs, Reed–Solomon codes, transparent SNARKs, post-quantum cryptography.

## 1 Introduction

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### 1.1 Background and Motivation

A *zero-knowledge proof* (ZKP) allows a computationally unbounded prover  $\mathcal{P}$  to convince a probabilistic polynomial-time verifier  $\mathcal{V}$  that a statement  $x \in L$  is true, without revealing *anything* about the witness  $w$  such that  $(x, w) \in R$  [GMR89]. The celebrated completeness–soundness–zero-knowledge trilemma makes designing efficient ZK proofs non-trivial, especially when the underlying NP relation  $R_L$  describes a large arithmetic circuit.

Classical constructions (e.g. Goldreich, Micali, Wigderson 1987 [GMW87]) reduce *any* NP relation to 3-Coloring via a Karp reduction and prove knowledge of the coloring. This approach is conceptually clean but expensive: communication grows as  $\mathcal{O}(k \cdot s)$ , where  $k$  is the security parameter and  $s$  is the circuit size. Crucially, prior techniques for converting semi-honest MPC protocols into malicious-secure ones (the “GMW compiler”) required *non-black-box* access to the underlying protocol — a serious limitation for both modularity and efficiency.

### 1.2 Main Contributions of the Two Papers

**IKOS07: Zero-Knowledge from Secure MPC.** Ishai, Kushilevitz, Ostrovsky, and Sahai [IKOS07] introduced the *MPC-in-the-head* paradigm: the prover internally simulates a multi-party computation among  $n$  virtual players, commits to each player’s view, and then opens only the challenged views. The key insight is that *zero-knowledge follows from*

the privacy of the underlying MPC protocol, while soundness follows from its robustness. This yields:

- The first black-box reduction from ZK proofs to semi-honest secure MPC.
- The first **constant-rate** ZK proof system: communication  $\mathcal{O}(s) + \text{poly}(k, \log s)$  for any NP relation with circuit size  $s$ .
- Near-witness-length ZK for constant-depth circuits:  $|w| \cdot \text{poly}(k) \cdot \text{polylog}(s)$ .

**AHIV22: Ligerio.** Ames, Hazay, Ishai, and Venkatasubramanian [AHIV22] gave a concrete, practically efficient instantiation of the MPC-in-the-head framework. Ligerio:

- Achieves **sublinear**  $\mathcal{O}(\sqrt{s})$  communication complexity.
- Requires only a **collision-resistant hash function** (CRHF) — no trusted setup, no public-key cryptography.
- Can be made **non-interactive** in the random oracle model via Fiat–Shamir, giving a transparent zk-SNARK.
- For SHA-256 preimage verification with  $2^{-40}$  soundness error, communication is approximately **35 KB**.

### 1.3 Organisation of This Report

Section 2 introduces the formal definitions needed throughout. Section 3 presents the IKOS07 paper: the MPC-in-the-head protocol, security analysis, and efficiency results. Section 4 presents Ligerio: the Damgård–Ishai MPC variant, interleaved Reed–Solomon codes, the zero-knowledge interactive PCP, and the main theorems. Section 5 compares the two papers. Section 6 discusses what the papers do not solve and lists open problems. Section 8 describes subsequent work.

## 2 Preliminaries and Definitions

Throughout,  $\lambda$  denotes the security parameter and  $\text{negl}(\lambda)$  denotes a negligible function. We write  $[n] = \{1, \dots, n\}$  and  $\binom{[n]}{t}$  for the set of all  $t$ -element subsets of  $[n]$ . All algorithms are uniform PPT unless stated otherwise.

### 2.1 Interactive Proof Systems and Zero-Knowledge

**Definition 2.1** (NP Relation). An NP relation is a polynomial-time decidable binary relation  $R \subseteq \{0, 1\}^* \times \{0, 1\}^*$  such that  $(x, w) \in R$  implies  $|w| \leq \text{poly}(|x|)$ . The associated language is  $L_R = \{x : \exists w, (x, w) \in R\}$ .

**Intuition:** Think of  $x$  as a problem instance (e.g. a SAT formula) and  $w$  as a solution (e.g. a satisfying assignment). The relation  $R$  simply checks that the solution is valid.

**Definition 2.2** (Interactive Proof System). An interactive proof system for  $L$  is a pair  $(\mathcal{P}, \mathcal{V})$  of interactive algorithms satisfying:

**Completeness:**  $\forall x \in L, \Pr[\langle \mathcal{P}(x, w), \mathcal{V}(x) \rangle = 1] \geq 1 - \text{negl}(\lambda)$ .

**Soundness:**  $\forall x \notin L, \forall \mathcal{P}^*, \Pr[\langle \mathcal{P}^*(x), \mathcal{V}(x) \rangle = 1] \leq \text{negl}(\lambda)$ .

**Intuition:** An honest prover can always convince the verifier of a true statement; no cheating prover can convince the verifier of a false statement with more than negligible probability.

**Definition 2.3** (Zero-Knowledge Proof). An interactive proof  $(\mathcal{P}, \mathcal{V})$  for  $L_R$  is (computational) zero-knowledge if for every PPT verifier  $\mathcal{V}^*$  there exists a PPT simulator  $\mathcal{S}$  such that for all  $(x, w) \in R$ ,

$$\{\mathcal{S}(x)\} \stackrel{c}{\approx} \{\text{View}_{\mathcal{V}^*}[\langle \mathcal{P}(x, w), \mathcal{V}^*(x) \rangle]\},$$

where  $\text{View}_{\mathcal{V}^*}$  denotes the transcript seen by  $\mathcal{V}^*$ . If the distributions are identical (resp. statistically close) we say the proof is *perfect* (resp. *statistical*) ZK.

**Intuition:** The verifier could have generated the entire transcript on its own (via the simulator  $\mathcal{S}$ ) without ever talking to the prover, so the interaction leaks no information about the witness  $w$ .

**Definition 2.4** (Proof of Knowledge).  $(\mathcal{P}, \mathcal{V})$  is a proof of knowledge for  $R$  with knowledge error  $\kappa$  if there is a PPT extractor  $\mathcal{E}$  such that for any  $\mathcal{P}^*$  with  $\Pr[\langle \mathcal{P}^*(x), \mathcal{V}(x) \rangle = 1] > \kappa$ , the extractor  $\mathcal{E}^{\mathcal{P}^*}(x)$  outputs  $w$  with  $(x, w) \in R$  in expected polynomial time.

**Intuition:** If a prover can convince the verifier, then there must be an extractor that can “rewind” the prover and pull out an actual witness. This goes beyond soundness: the prover doesn’t just fail to lie, it genuinely knows the secret.

## 2.2 Commitment Schemes

**Definition 2.5** (Commitment Scheme). A commitment scheme  $\text{Com}$  consists of algorithms  $(\text{Setup}, \text{Commit}, \text{Open})$  satisfying:

**Binding:** No PPT adversary can open a commitment to two different values.

**Hiding:**  $\text{Com}(m_0; r) \stackrel{c}{\approx} \text{Com}(m_1; r)$  for all  $m_0, m_1$ .

A commitment is *statistically binding* if binding holds against computationally unbounded adversaries. A hash-based commitment  $\text{Com}(m; r) = H(m||r)$  is computationally hiding in the random oracle model and computationally binding assuming collision resistance of  $H$ .

**Intuition:** Think of putting a sealed envelope on the table: the sender cannot change what is inside (binding), and the receiver cannot see the contents until the envelope is opened (hiding). The commitment is the envelope; decommitting is opening it.

## 2.3 Secret Sharing

**Definition 2.6** ( $t$ -out-of- $n$  Secret Sharing). A  $t$ -out-of- $n$  secret sharing scheme  $(\text{Share}, \text{Rec})$  for a secret  $s \in \mathcal{S}$  produces shares  $(s_1, \dots, s_n)$  such that:

- (*Privacy*) Any set of at most  $t$  shares reveals nothing about  $s$ .
- (*Correctness*) Any set of  $t + 1$  or more shares uniquely determines  $s$ .

**Intuition:** A secret is broken into  $n$  pieces so that any small group of conspirators ( $\leq t$  of them) learns nothing, while any sufficiently large group ( $\geq t + 1$ ) can reconstruct the secret.

**Example 2.7** (XOR Sharing). For  $s \in \{0, 1\}^m$ , pick  $s_1, \dots, s_{n-1} \leftarrow_R \{0, 1\}^m$  and set  $s_n = s \oplus s_1 \oplus \dots \oplus s_{n-1}$ . This is an  $n$ -out-of- $n$  perfect secret sharing scheme with privacy threshold  $t = n - 1$ . Any  $n - 1$  shares are uniformly random and independent of  $s$ .

**Example 2.8** (Shamir Secret Sharing [Shamir79]). Over a finite field  $\mathbb{F}_q$ , choose a polynomial  $f$  of degree  $\leq t$  uniformly at random with  $f(0) = s$  and set  $s_i = f(\alpha_i)$  for distinct  $\alpha_i \in \mathbb{F}_q^*$ . Any  $t$  shares are perfectly private; any  $t + 1$  shares uniquely reconstruct  $s$  by Lagrange interpolation.

## 2.4 Secure Multiparty Computation

**Definition 2.9** ( $t$ -Private MPC (Semi-Honest)). A protocol  $\Pi$  among  $n$  players computing functionality  $f : (\{0, 1\}^*)^n \rightarrow \{0, 1\}^*$  is  $t$ -private if there is a PPT simulator  $\mathcal{S}$  such that for every set  $T \subseteq [n]$  with  $|T| \leq t$  and every input tuple  $(x_1, \dots, x_n)$ ,

$$\{\mathcal{S}(T, \{x_i\}_{i \in T}, f(x_1, \dots, x_n))\} \equiv \{\text{View}_T[\Pi(x_1, \dots, x_n)]\},$$

where  $\text{View}_T$  is the joint view of the players in  $T$  (inputs, random coins, messages received).

**Intuition:** Even if up to  $t$  curious-but-honest players pool their views, they learn nothing about the other players' inputs beyond what the output itself reveals.

**Definition 2.10** ( $t$ -Robust MPC (Malicious)).  $\Pi$  is *statistically  $t$ -robust* if even when  $\leq t$  players behave arbitrarily, the outputs of the honest players are correct with overwhelming probability. Formally, for any adversary  $\mathcal{A}$  controlling  $\leq t$  players, the output of each honest player equals  $f(x_1, \dots, x_n)$  except with negligible probability.

**Intuition:** Even actively malicious parties that deviate arbitrarily from the protocol cannot corrupt the outputs of honest players. Robustness is a stronger guarantee than privacy: the protocol not only hides inputs but guarantees correct outputs.

*Remark 2.11.* The Ben-Or–Goldwasser–Wigderson (BGW) protocol [BGW88] achieves information-theoretic  $t$ -privacy and  $t$ -robustness for  $t < n/2$  and  $t < n/3$ , respectively, over any functionality computable by an arithmetic circuit.

## 2.5 Coin-Flipping Protocol

**Definition 2.12** (Coin-Flipping Protocol). A two-party coin-flipping protocol between  $\mathcal{P}$  and  $\mathcal{V}$  produces shared random coins  $r^* \in \{0, 1\}^\ell$  that are:

- *Uniform:*  $r^*$  is uniformly distributed (No polynomial-time party can bias it by more than negligible probability).
- *Binding:* Once committed, neither party can change  $r^*$ .

**Intuition:** Neither party should be able to choose the randomness to its advantage. The standard construction has each party commit to their own random string, then XOR the two strings together: even if one party cheats and picks its string after seeing the other's commitment, the XOR is still uniform because the honest party's string is truly random.

**Why is coin-flipping needed in  $\Pi'_R$ ?** The basic protocol  $\Pi_R$  assumes the MPC protocol  $\Pi_f$  is *perfectly correct*: on a valid witness, every execution produces output 1. Some MPC protocols are only *statistically correct*: they fail with small probability that depends on the internal randomness. If the prover chooses the randomness  $r$  for  $\Pi_f$  on its own, a dishonest prover could pick  $r$  to be a “bad” value where the protocol happens to produce output 1 even when  $x \notin L$ . The coin-flipping preamble forces the prover and verifier to jointly and verifiably sample the randomness  $r^*$  for  $\Pi_f$ , so neither party can manipulate it.

## 2.6 Error-Correcting Codes

**Definition 2.13** (Linear Code). A linear code  $\mathcal{C}$  with parameters  $[m, k, d]_q$  over  $\mathbb{F}_q$  is a  $k$ -dimensional subspace of  $\mathbb{F}_q^m$  where every nonzero codeword has Hamming weight  $\geq d$  (the minimum distance). The relative distance is  $\delta = d/m$ .

**Intuition:** Encoding adds redundancy so that even if some symbols are corrupted (or withheld), one can detect (or correct) the errors. The minimum distance  $d$  measures how many positions must differ between any two distinct codewords — a larger  $d$  means the code can tolerate more errors.

**Definition 2.14** (Reed–Solomon Code). The Reed–Solomon code  $\text{RS}_q[m, k]$  consists of all evaluations  $(f(\alpha_1), \dots, f(\alpha_m))$  of polynomials  $f$  of degree  $< k$  over  $\mathbb{F}_q$ , where  $\alpha_1, \dots, \alpha_m$  are distinct field elements. It has parameters  $[m, k, m - k + 1]_q$ , achieving the Singleton bound (MDS code).

**Intuition:** A degree- $(k-1)$  polynomial is determined by  $k$  points; evaluating it at  $m > k$  points creates redundancy. Two distinct polynomials of degree  $< k$  agree on at most  $k - 1$  points, so their evaluations differ in at least  $m - k + 1$  positions. This large minimum distance makes RS codes ideal for detecting inconsistencies in the prover’s claimed encodings.

**Definition 2.15** (Algebraic-Geometric (AG) Code). AG codes generalise Reed–Solomon codes using algebraic curves. For a curve  $\mathcal{X}$  over  $\mathbb{F}_q$  of genus  $g$  with  $N$  rational points, one obtains codes with parameters  $[N, k, d]_q$  where  $d \geq N - k - (2g - 2)$ . When  $g$  is small relative to  $k$ , these approach the Singleton bound [GS95].

**Intuition:** AG codes allow one to construct long codes over small fields (e.g.  $\mathbb{F}_2$ ) with good parameters, which is not possible with plain RS codes. IKOS07 uses AG codes to build the constant-rate ZK proof over circuits of arbitrary depth.

## 2.7 Interactive PCP and IOP

**Definition 2.16** (Interactive PCP (ZKIPCP)). An interactive PCP consists of a prover that writes a PCP string  $\pi$  and then interacts with the verifier, who may query positions of  $\pi$  adaptively. A zero-knowledge ZKIPCP additionally hides the witness from the verifier.

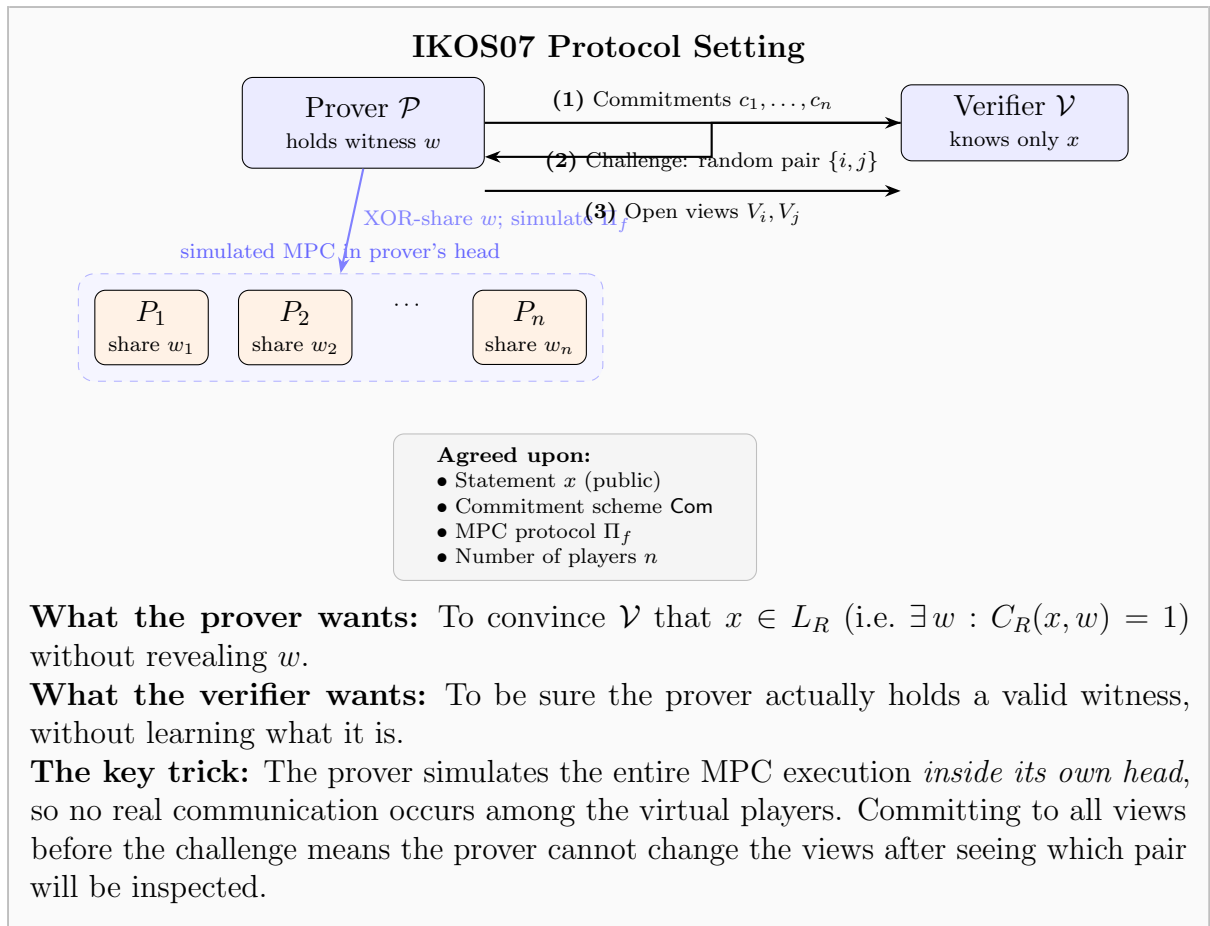
**Intuition:** The prover publishes an oracle (an encrypted book of all the computation), and the verifier checks a small random sample. The interaction allows the verifier to ask follow-up questions adaptively. Ligero’s protocol is precisely such a ZKIPCP whose “book” is the interleaved RS encoding of the witness.

**Definition 2.17** (Proximity Testing / IOPP). Given query access to a function  $f : S \rightarrow \mathbb{F}_q$ , a proximity test verifies whether  $f$  is  $\delta$ -close to a codeword  $c \in \mathcal{C}$  with high probability using far fewer than  $|S|$  queries.

**Intuition:** Rather than checking every symbol of the alleged codeword, the verifier queries a small random sample. If the word is far from all codewords, the random sample will witness a discrepancy with high probability. This is how Ligero achieves sublinear communication: the verifier only queries  $\mathcal{O}(\sqrt{s})$  columns of the  $\sqrt{s} \times \sqrt{s}$  witness matrix.

### 3 Zero-Knowledge from Secure MPC (IKOS07)

#### 3.1 Setting and High-Level Idea



#### 3.2 Overview and Goals

The central result of [IKOS07] is:

**Theorem 3.1** (Ishai–Kushilevitz–Ostrovsky–Sahai, Informal). *If there exists an  $n$ -party protocol  $\Pi_f$  computing  $f$  with  $t$ -privacy and  $t$ -robustness, then there is a zero-knowledge proof system for any NP relation  $R$  with communication complexity  $\mathcal{O}(s) + \text{poly}(k, \log s)$ , where  $s = |C_R|$  is the circuit size of  $R$ .*

The reduction is *black-box*: it uses  $\Pi_f$  only as an oracle, never looking inside its code.



### 3.3 The Functionality and MPC Protocol Setup

Given an NP relation  $R$  with circuit  $C_R$  of size  $s$ , define the  $(n + 1)$ -party functionality

$$f_R(x; w_1, \dots, w_n) = C_R(x, w_1 \oplus w_2 \oplus \dots \oplus w_n),$$

where  $x$  is a public input (the statement) and each  $w_i$  is a private input (a share of the witness  $w$ ). The prover secret-shares  $w$  among  $n$  virtual players using XOR sharing, so  $w_1 \oplus \dots \oplus w_n = w$ .

**Definition 3.1** (Player View). The *view*  $V_i$  of player  $P_i$  in an execution of  $\Pi_f$  consists of: (i)  $P_i$ 's private input  $w_i$ , (ii)  $P_i$ 's internal random coins  $r_i$ , and (iii) all messages received by  $P_i$  from other players.

**Definition 3.2** (Consistent Views). Two views  $V_i$  and  $V_j$  are *consistent* with respect to  $\Pi_f$  and public input  $x$  if the messages that  $P_i$  sent to  $P_j$  (recorded in  $V_j$ ) match what  $P_i$  would produce given  $V_i$ , and vice versa.

**Intuition:** Two views are consistent if they tell the same story from two different vantage points. If both players agree on all the messages they exchanged, their local records are consistent.

### 3.4 The Basic Protocol $\Pi_R$

#### Protocol $\Pi_R$ — Basic MPC-in-the-Head Construction

**Common input:** Statement  $x$ . **Prover's private input:** Witness  $w$ ,  $C_R(x, w) = 1$ .

**Parameters:**  $n$  players, commitment scheme  $\text{Com}$ .

1. **(Secret-Share & Simulate).** Prover picks  $w_1, \dots, w_{n-1} \leftarrow_R \{0, 1\}^{|w|}$ , sets  $w_n = w \oplus w_1 \oplus \dots \oplus w_{n-1}$ , and runs  $\Pi_f(x; w_1, \dots, w_n)$  internally with fresh random coins  $r_1, \dots, r_n$ , obtaining views  $V_1, \dots, V_n$ .
2. **(Commit).** Prover sends commitments  $c_i = \text{Com}(V_i; \rho_i)$  for  $i \in [n]$ .
3. **(Challenge).** Verifier picks a uniformly random pair  $\{i, j\} \in \binom{[n]}{2}$  and sends it.
4. **(Open).** Prover sends  $V_i, V_j, \rho_i, \rho_j$ .
5. **(Verify).** Verifier accepts iff: (a) commitments verify; (b) outputs of  $P_i, P_j$  are both 1; (c)  $V_i$  and  $V_j$  are consistent w.r.t.  $\Pi_f$  and  $x$ .

**Soundness amplification:** Repeat  $\ell = \lceil k \cdot \binom{n}{2} \rceil$  times independently; accept iff all rounds accept.

### 3.5 Security Analysis of $\Pi_R$

#### 3.5.1 Completeness

If  $x \in L_R$  and the prover holds a valid witness, the simulated execution is honest: every pair  $(V_i, V_j)$  is consistent and produces output 1. **Perfect completeness holds.**

### 3.5.2 Soundness

**Lemma 3.2** (Local-Global Consistency, Ishai–Kushilevitz–Ostrovsky–Sahai [IKOS07]). *The  $n$  views  $V_1, \dots, V_n$  correspond to an honest execution of  $\Pi_f$  on some inputs if and only if every pair  $(V_i, V_j)$  is pairwise consistent.*

*Proof.* The forward direction is immediate: in a valid execution all pairwise interactions are consistent by definition. For the converse, suppose every pair is pairwise consistent. Fix any player  $P_i$ . The messages in  $V_i$  alleged to have been sent by  $P_j$  must equal what  $P_j$  would produce given  $V_j$  (by consistency of  $(V_i, V_j)$ ). Since  $j$  is arbitrary, the collection  $V_1, \dots, V_n$  forms a valid global transcript for  $\Pi_f$  on some input tuple.

**Intuition:** *Pairwise consistency is equivalent to global consistency: if every pair of players tells a coherent local story, there must be a single coherent global story. This reduces checking a complex global constraint to checking  $\binom{n}{2}$  simple pairwise constraints.*

**Lemma 3.3** (Soundness of  $\Pi_R$ ). *If  $x \notin L_R$ , the acceptance probability of a single round of  $\Pi_R$  is at most  $1 - 1/\binom{n}{2}$ .*

*Proof.* Suppose  $x \notin L_R$ . Since  $\Pi_f$  is perfectly correct, every valid execution on any input tuple produces output 0. If all committed views are globally consistent, step 5(b) fails. If some pair  $(V_{i^*}, V_{j^*})$  is inconsistent, the verifier’s uniform challenge equals  $\{i^*, j^*\}$  with probability  $\geq 1/\binom{n}{2}$ , making step 5(c) fail. In either case, acceptance probability  $\leq 1 - 1/\binom{n}{2}$ .

**Corollary 3.4.** *After  $\ell = k \cdot \binom{n}{2}$  repetitions, soundness error  $(1 - 1/\binom{n}{2})^\ell \leq e^{-k} = \text{negl}(k)$ .*

### 3.5.3 Zero-Knowledge

**Theorem 3.5** (ZK of  $\Pi_R$ ). *If  $\Pi_f$  is 2-private, then  $\Pi_R$  is computationally zero-knowledge assuming  $\text{Com}$  is computationally hiding.*

*Proof Sketch.* Simulator  $\mathcal{S}$  on input  $x$ :

1. Run  $\mathcal{V}^*(x)$  to get its challenge  $\{i, j\}$  (or guess uniformly).
2. Sample  $w_i, w_j \leftarrow_R \{0, 1\}^{|w|}$  independently.
3. Run the MPC privacy simulator  $\mathcal{S}_{\Pi_f}(\{i, j\}, x, w_i, w_j, 1)$  to get simulated views  $\tilde{V}_i, \tilde{V}_j$ .
4. For  $h \notin \{i, j\}$ , set  $\tilde{V}_h$  to random strings.
5. Commit:  $\tilde{c}_h = \text{Com}(\tilde{V}_h; \rho_h)$  and output the full transcript.

*Indistinguishability:* In the real execution, shares  $w_i, w_j$  are uniform and independent (XOR sharing). By 2-privacy of  $\Pi_f$ , the joint view  $(V_i, V_j)$  is identically distributed to  $\mathcal{S}_{\Pi_f}(\{i, j\}, x, w_i, w_j, 1)$ . The hiding property of  $\text{Com}$  ensures that the  $n - 2$  unopened commitments are computationally indistinguishable from random commitments to the actual views.

If the simulator guesses the challenge wrong ( $\binom{n}{2} - 1$  out of  $\binom{n}{2}$  cases), it outputs  $\perp$  and restarts; the expected number of restarts is  $\binom{n}{2}$ . Hence  $\mathcal{S}$  runs in expected polynomial time.

### 3.6 Protocol $\Pi'_R$ : Handling Statistical Correctness

For MPC protocols with only *statistical* (rather than perfect) correctness, a coin-flipping preamble is added before the main protocol.

#### Protocol $\Pi'_R$ — Statistical Correctness Variant

**Preamble (Coin-Flipping):**  $\mathcal{P}$  and  $\mathcal{V}$  jointly execute a coin-flipping protocol to agree on shared randomness  $r^*$ . The prover then uses  $r^*$  as the internal randomness for  $\Pi_f$ .

- The verifier can later check that  $\Pi_f$  was run with exactly  $r^*$  (binding of the coin flip prevents the prover from substituting a “lucky”  $r$  after seeing the challenge).
- Because  $r^*$  is chosen jointly, neither party can bias the randomness to force a statistical failure.

**Main protocol:** Identical to  $\Pi_R$  using the agreed  $r^*$ .

**Intuition:** Without the coin-flip preamble, a cheating prover could try exponentially many random tapes until it finds one where the statistically incorrect MPC happens to accept a false statement. The coin-flip forces the random tape to be chosen before the prover commits, closing this loophole.

### 3.7 Protocol $\Pi_{R,t}$ : Robust MPC and Single-Round Amplification

The basic protocol requires  $\mathcal{O}(kn^2)$  repetitions for soundness amplification. Using a  $t$ -robust MPC reduces repetitions dramatically.

#### Protocol $\Pi_{R,t}$ — Robust MPC Variant

Same as  $\Pi_R$  but:

1. Use  $\Pi_f$  that is  $t$ -private and  $t$ -robust with  $n = ct$  players.
2. Verifier chooses a random set  $T \subseteq [n]$ ,  $|T| = t$ .
3. Prover opens all views  $\{V_i\}_{i \in T}$ .
4. Verifier checks all  $\binom{t}{2}$  pairs in  $T$  for consistency and all outputs equal 1.

**Theorem 3.6** (Soundness of  $\Pi_{R,t}$ , Ishai–Kushilevitz–Ostrovsky–Sahai). *If  $x \notin L_R$  and  $\Pi_f$  is  $t$ -robust, then  $\Pi_{R,t}$  accepts with probability at most  $2^{-\Omega(k)}$ .*

*Proof Sketch.* Let  $G = ([n], E)$  be the inconsistency graph where  $(i, j) \in E$  iff  $V_i$  and  $V_j$  are inconsistent.

**Case 1:**  $\text{min-vertex-cover}(G) \leq t$ . Let  $S^*$  be the vertex cover. Views outside  $S^*$  are mutually consistent. By  $t$ -robustness,  $\geq n - t$  players output 0. The random set  $T$  hits one with probability  $\geq 1 - 2^{-\Omega(k)}$ .

**Case 2:**  $\text{min-vertex-cover}(G) > t$ . By König’s theorem,  $G$  has a matching of size  $> t/2$ , giving  $> t/2$  disjoint inconsistent pairs. The random set  $T$  of size  $t$  hits at least one edge of the matching with probability  $\geq 1 - 2^{-\Omega(k)}$ .

**Intuition:** Opening  $t$  views instead of 2 is much harder to fool: a cheating prover must hide all inconsistencies from a large random set, which is exponentially unlikely when many inconsistencies exist.

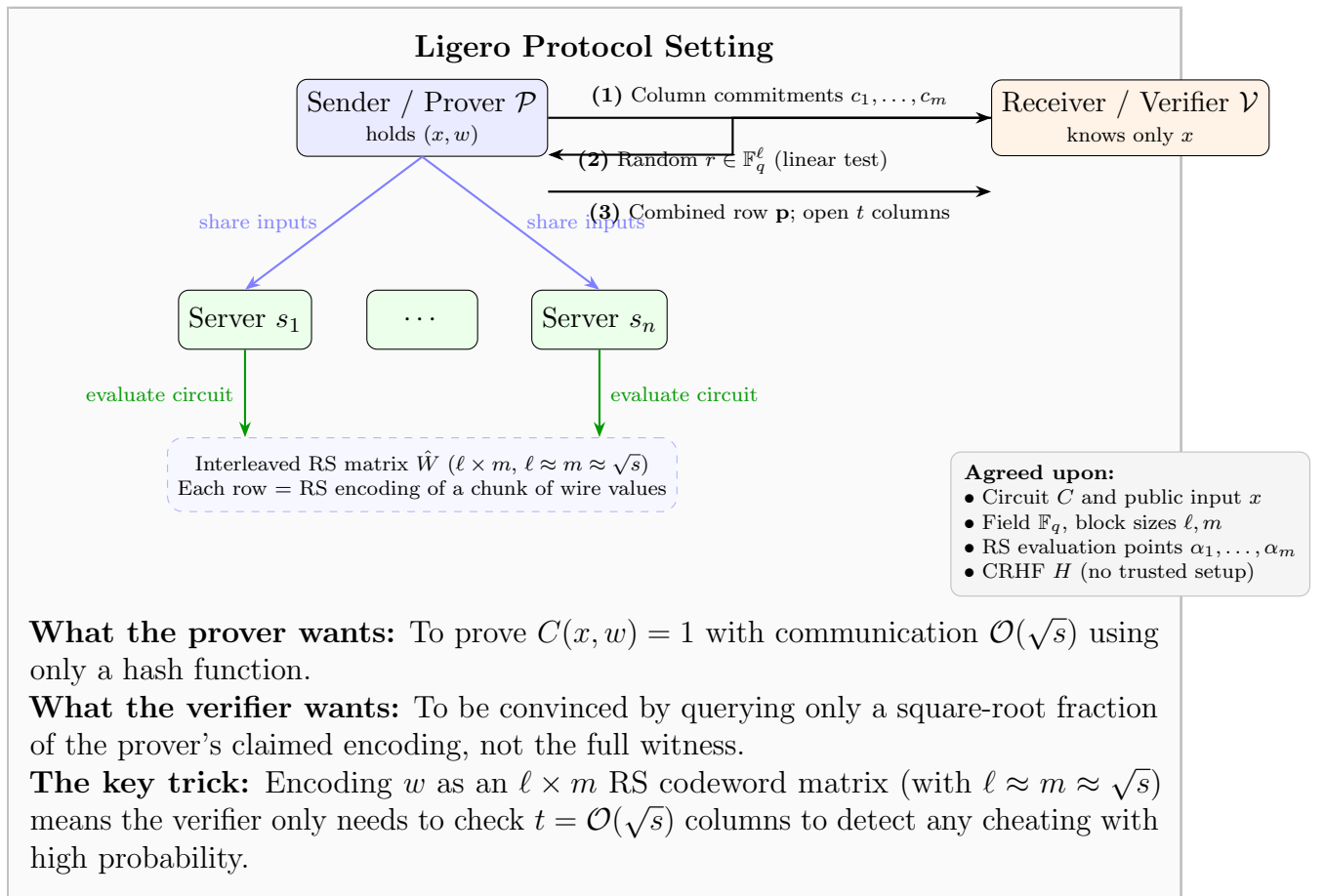
### 3.8 Efficiency Analysis

**Theorem 3.7** (Communication Complexity of IKOS07). *Using the Damgård–Ishai MPC protocol [DI06] and algebraic-geometry codes [CC06], the resulting ZK proof for any  $L_R$  with circuit size  $s$  has communication complexity  $\mathcal{O}(s) + \text{poly}(k, \log s)$ , the first **constant-rate** ZK proof for all of NP based on one-way functions.*

For constant-depth circuits ( $\text{NC}^0$  or  $\text{NC}^1$ ) the protocol further achieves near-witness-length:  $|w| \cdot \text{poly}(k) \cdot \text{polylog}(s)$ .

## 4 Ligerio: Lightweight Sublinear ZK Arguments

### 4.1 Setting and High-Level Idea



### 4.2 Overview and Goals

While IKOS07 achieves constant-rate (linear communication) for all NP, the constant in  $\mathcal{O}(s)$  is large in practice. Ligerio [AHIV22] pursues *sublinear* communication  $\mathcal{O}(\sqrt{s})$  at the cost of *computational* (rather than statistical) soundness.

### 4.3 The Damgård–Ishai MPC Protocol

Ligerio uses an optimised variant of the Damgård–Ishai protocol [DI06]: a sender  $S$  with input  $(x, w)$  secret-shares inputs to  $n$  servers, which evaluate the arithmetic circuit gate by

gate and deliver the output to a receiver  $R$  (the verifier). The protocol achieves  $t_p$ -privacy against passive adversaries and  $t_r$ -robustness against malicious adversaries.

**Definition 4.1** (Statistical  $t_r$ -Robustness).  $\Pi$  realises  $f$  with statistical  $t_r$ -robustness if (a) it is perfectly correct against honest-but-curious adversaries, and (b) even when  $\leq t_r$  servers behave maliciously, the receiver’s output equals  $f(x)$  except with negligible probability.

#### 4.4 Interleaved Reed–Solomon Codes

**Definition 4.2** (Interleaved RS Encoding). Let  $k = \lceil s/\ell \rceil$  for block length  $\ell$ . Write the wire-value vector  $w^* \in \mathbb{F}_q^s$  as  $\ell$  rows  $u^{(1)}, \dots, u^{(\ell)} \in \mathbb{F}_q^k$ . The *interleaved* RS encoding is:

$$\hat{W} = \begin{pmatrix} \text{enc}(u^{(1)}) \\ \vdots \\ \text{enc}(u^{(\ell)}) \end{pmatrix} \in \mathbb{F}_q^{\ell \times m}, \quad \ell \approx m \approx \sqrt{s}.$$

**Intuition:** Rather than committing to the witness as a flat vector (which would cost  $s$  field elements), we arrange it as a near-square matrix and commit to each column. The verifier then checks  $\mathcal{O}(\sqrt{s})$  columns instead of  $\mathcal{O}(s)$  positions, giving sublinear communication.

**Lemma 4.1** (Proximity of Interleaved RS, Ames–Hazay–Ishai–Venkatasubramanian [AHIV22]). Let  $\hat{W}$  be  $\delta_0$ -far from the interleaved RS code (i.e.,  $\geq \delta_0 m$  columns are not RS codewords). Then for a uniformly random linear combination  $r \in \mathbb{F}_q^\ell$ , the vector  $r^\top \hat{W}$  is  $\delta_0(1 - \delta_0)$ -far from  $\text{RS}_q[m, k]$  except with probability  $\ell/q$ .

*Proof Sketch.* For a bad column  $j$ , the linear combination  $r^\top \hat{W}_j$  is non-zero for a uniformly random  $r$  with probability  $\geq 1 - 1/q$  (Schwartz–Zippel lemma). A union bound over all  $\geq \delta_0 m$  bad columns gives the result.

**Intuition:** A random linear combination of the rows “mixes” the bad columns into the combined row, making the combined row far from any RS codeword. This lets the verifier detect encoding failures by checking just one combined row instead of all  $\ell$  rows separately.

#### 4.5 The Ligero ZK Protocol

The circuit  $C_R$  is expressed as an arithmetic circuit over  $\mathbb{F}_q$  (with  $q > s$ ). Denote the wire-value vector by  $w^* \in \mathbb{F}_q^s$ . Each multiplication gate  $g_i : a \cdot b = c$  imposes:

$$\mathbf{a}_i^\top w^* \cdot \mathbf{b}_i^\top w^* = \mathbf{c}_i^\top w^*, \quad i \in [s].$$

##### Ligero Zero-Knowledge Protocol (Simplified)

**Common input:** Circuit  $C$  and statement  $x$ . **Prover’s input:** Witness  $w$  with  $C(x, w) = 1$ .

1. **(Encode and Commit).** Prover forms the  $\ell \times m$  interleaved RS matrix  $\hat{W}$  from the wire-value vector  $w^*$ . Commits to each column  $j$ :  $c_j = H(\hat{W}_{*,j})$ .
2. **(Linear Test — Row Checks).** Verifier sends random  $r \in \mathbb{F}_q^\ell$ . Prover sends combined row  $\mathbf{p} = r^\top \hat{W}$ . Verifier picks  $t$  random columns; prover opens them. Verifier checks each opened column is consistent with its commitment and with

**p.**

3. **(Quadratic Test — Gate Checks).** The constraints are mapped to an  $\ell \times k$  grid. The verifier sends a random vector  $\mathbf{r}_{gate} \in \mathbb{F}_q^\ell$ . The prover linearly combines the constraint rows using  $\mathbf{r}_{gate}$  and sends summary polynomials  $\mathbf{p}_a, \mathbf{p}_b, \mathbf{p}_c$  of size  $\mathcal{O}(\sqrt{s})$ . The verifier checks  $\mathbf{p}_a \cdot \mathbf{p}_b = \mathbf{p}_c$  and opens  $t$  more columns.
4. **(Accept).** Verifier accepts iff all linear and quadratic checks pass.

#### 4.6 Security of Ligero

**Theorem 4.2** (Completeness of Ligero). *If  $C(x, w) = 1$  and the prover is honest, the verifier always accepts.*

*Proof.* All gate constraints are satisfied by construction; the linear and quadratic checks follow from the circuit's arithmetic; all column openings are consistent with the CRHF commitments.

**Theorem 4.3** (Soundness of Ligero, Ames–Hazay–Ishai–Venkatasubramanian). *If  $C(x, w) = 0$  for all  $w$ , the verifier accepts with probability at most  $\mathcal{O}(\sqrt{s}/q) = \text{negl}(k)$  for  $q = \text{poly}(s, k)$ .*

*Proof Sketch.* A cheating prover must either (a) commit to an encoding  $\hat{W}$  that is far from the interleaved RS code, or (b) commit to a valid encoding that does not satisfy the gate constraints.

**Case (a) — Encoding check:** By Lemma 4.1,  $r^\top \hat{W}$  is far from  $\text{RS}[m, k]$  with high probability; opening  $t$  columns reveals a discrepancy.

**Case (b) — Constraint check:** At least one gate constraint fails. The verifier's random linear combination  $\mathbf{r}_{gate} \in \mathbb{F}_q^\ell$  over the constraint rows will fail to cancel out the error with probability  $1 - 1/q$  (by the Schwartz–Zippel lemma).

Setting  $\ell = t = \sqrt{s}$  and  $q = \text{poly}(s, k)$  gives negligible soundness.

**Theorem 4.4** (Zero-Knowledge of Ligero). *Ligero is honest-verifier zero-knowledge. It is made fully ZK by adding a  $\Sigma$ -protocol for the committed values, at a constant factor in communication.*

*Proof Sketch.* Ligero achieves ZK by adding a "blinding row" of uniformly random field elements to the matrix  $\hat{W}$  before applying the RS encoding. When the verifier requests the linear combination  $\mathbf{p} = r^\top \hat{W}$ , the inclusion of this random row acts as a one-time pad, ensuring  $\mathbf{p}$  is uniformly distributed and leaks nothing about the witness. The computational ZK for the unopened columns follows from the hiding property of the CRHF commitments.

#### 4.7 Communication Complexity of Ligero

**Theorem 4.5** (Sublinear Communication, Ames–Hazay–Ishai–Venkatasubramanian). *The communication complexity of Ligero for a circuit  $C$  of size  $s$  is  $\mathcal{O}(\sqrt{s} \cdot \lambda)$  bits, where  $\lambda$  is the security parameter.*

*Proof.* The encoding  $\hat{W}$  is an  $\ell \times m$  matrix where  $\ell \approx m \approx \sqrt{s}$ . Committing to all  $m$  columns using a Merkle tree requires sending just the root hash ( $\lambda$  bits). The verifier's

linear combination vector  $r$  and the prover’s combined row response  $\mathbf{p}$  together contribute  $\mathcal{O}(\ell + m) = \mathcal{O}(\sqrt{s})$  field elements.

To achieve a soundness error of  $2^{-\lambda}$ , the verifier opens a security-parameter-dependent number of columns,  $t = \Theta(\lambda)$ . Opening  $t$  columns requires sending  $t \cdot \ell$  field elements plus their Merkle authentication paths. The communication for these openings is  $\mathcal{O}(t \cdot \ell + t \cdot \lambda \log m)$ . Substituting  $\ell \approx \sqrt{s}$  and  $t \approx \lambda$ , the dominant term becomes  $\mathcal{O}(\lambda\sqrt{s})$ . Thus, the total communication is strictly sublinear in the circuit size.

Table 1: Concrete communication for SHA-256 preimage,  $2^{-40}$  soundness error

| Scheme                                               | Communication                             | Assumptions |
|------------------------------------------------------|-------------------------------------------|-------------|
| GMW (Goldreich–Micali–Wigderson)                     | $\mathcal{O}(k \cdot s)$                  | OWFs        |
| IKOS07 (constant-rate)                               | $\mathcal{O}(s) + \text{poly}(k)$         | OWFs        |
| ZKBoo (Giacomelli–Madsen–Orlandi)                    | $\approx 289 \text{ KB}$                  | CRHF’s      |
| ZKB++ (Chase et al.)                                 | $\approx 202 \text{ KB}$                  | CRHF’s      |
| <b>Ligerio (Ames–Hazay–Ishai–Venkatasubramaniam)</b> | <b><math>\approx 35 \text{ KB}</math></b> | CRHF’s      |

## 5 Comparison: IKOS07 vs. Ligerio

Table 2: High-level comparison of IKOS07 and Ligerio

| Property                 | IKOS07                                    | Ligerio (AHIV22)                      |
|--------------------------|-------------------------------------------|---------------------------------------|
| Communication            | $\mathcal{O}(s) + \text{poly}(k, \log s)$ | $\mathcal{O}(\sqrt{s} \cdot \lambda)$ |
| Soundness type           | Statistical (ZK proofs)                   | Computational (ZK arguments)          |
| Cryptographic assumption | OWFs                                      | CRHF’s only                           |
| Trusted setup            | None                                      | None                                  |
| Non-interactive          | No (unless Fiat–Shamir)                   | No (ROM via Fiat–Shamir)              |
| Black-box MPC use        | Yes (any semi-honest MPC)                 | No (specific Damgård–Ishai)           |
| Round complexity         | $\mathcal{O}(kn^2)$ or $\mathcal{O}(k)$   | $\mathcal{O}(1)$                      |
| Primary goal             | Theoretical unification                   | Practical efficiency                  |
| Key technique            | MPC-in-the-head, XOR sharing              | Interleaved RS codes, proximity       |
| Post-quantum             | Depends on OWF choice                     | Yes (CRHF = hash function)            |

**Theoretical vs. Practical Impact.** IKOS07 is a theoretical landmark: it establishes the first general black-box connection between ZK and MPC and achieves constant-rate



proofs. However, the constant in  $\mathcal{O}(s)$  is large in practice due to the Damgård–Ishai overhead and the complexity of algebraic-geometry code encoders.

Ligerio sacrifices modularity for efficiency: it uses a specific MPC and a simpler coding scheme (interleaved RS instead of AG codes), but achieves concretely small proofs of  $\approx 35$  KB for SHA-256.

**Modular vs. Specialised.** IKOS07’s black-box reduction means that any advance in MPC directly improves the ZK proof. Ligerio is tied to a specific construction, but this specificity enables the sublinear  $\mathcal{O}(\sqrt{s})$  bound.

## 6 What the Papers Do Not Solve

### 6.1 Limitations of IKOS07

- (a) **Practical efficiency.** The constant in  $\mathcal{O}(s)$  is large. Algebraic-geometry codes have complex encoders; the Damgård–Ishai protocol has significant  $\text{poly}(k, n)$  overhead. The protocol is primarily of theoretical interest without substantial implementation optimisation.
- (b) **Round complexity.** The basic protocol requires  $\mathcal{O}(kn^2)$  rounds. The advanced variant uses  $\mathcal{O}(k)$  rounds. *Constant-round ZK proofs under OWFs remain open.*
- (c) **Proof of knowledge.** The construction satisfies proof of knowledge informally but extractors are not explicitly constructed or optimised.
- (d) **Non-black-box separation.** The paper shows semi-honest  $\not\rightarrow$  malicious in a black-box way for the *same* functionality, but the exact black-box separations are not fully characterised.
- (e) **Adaptive corruptions.** General adaptive adversaries in the ZK setting are not addressed.
- (f) **Communication lower bounds.** Matching  $\Omega(s)$  lower bounds for ZK *proofs* (not arguments) are not established.

### 6.2 Limitations of Ligerio

- (a) **Computational soundness only.** Ligerio is an *argument* (computationally sound) rather than a *proof* (statistically sound). Against computationally unbounded provers, soundness fails.
- (b) **Verifier time.** The verifier must evaluate  $\mathcal{O}(\sqrt{s})$  hash evaluations; for very large circuits this may be slow.
- (c) **Prover time.** The prover performs  $\mathcal{O}(s)$  field operations plus  $\mathcal{O}(m)$  RS encodings, which can be expensive for very large circuits compared to linear-prover-time successors (e.g., Aurora [BCR19]).
- (d) **Proof size vs. subsequent work.** Later systems (e.g., Aurora [BCR19], STARK [BBHR18]) achieve polylogarithmic communication at the cost of more complex algebraic tools.



- (e) **Non-linear constraints.** Ligero handles only arithmetic circuits; supporting more general constraint systems (R1CS, PLONKish) requires non-trivial extensions.

## 7 Open Problems

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- Q1. Constant-rate ZK with constant rounds.** Can we achieve  $\mathcal{O}(s)$  communication and  $\mathcal{O}(1)$  rounds simultaneously under OWFs? IKOS07’s constant-rate protocol requires a  $\Theta(k)$ -round preamble.
- Q2. Optimal communication for ZK proofs.** Is  $\Omega(s)$  a lower bound for ZK *proofs* (statistical soundness) for arbitrary NP? Goldreich and Håstad [GH98] show that sublinear ZK proofs would imply surprising results for SAT.
- Q3. Black-box constructions of general MPC from semi-honest MPC.** IKOS07 shows a special case for ZK. Can one build fully malicious-secure MPC from semi-honest MPC in a black-box way for *all* functionalities?
- Q4. Sublinear-prover transparent ZK.** Ligero achieves  $\mathcal{O}(\sqrt{s})$  communication but  $\mathcal{O}(s)$  prover time. Can we achieve  $o(s)$  prover time while maintaining transparency (no trusted setup)?
- Q5. Optimal parameters for MPC-in-the-head.** For Ligero and its successors, what are the optimal choices of  $\ell$ ,  $m$ , and field size  $q$  to minimise total proof size in practice?
- Q6. VOLE-in-the-head and public verifiability.** Recent work [BBD23] shows that vector oblivious linear evaluation (VOLE) can replace MPC for certain ZK constructions, giving linear communication. Can VOLE-based proofs match Ligero’s sublinear communication?
- Q7. Better understanding of ZK  $\leftrightarrow$  MPC duality.** The paper notes ZK proofs implement a ZK variant of PCPs over large alphabets. A unified complexity-theoretic framework for PCPs, interactive proofs, and MPC in this context is lacking.

## 8 Subsequent Work and Impact

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**ZKBoo and ZKB++** (Giacomelli–Madsen–Orlandi 2016; Chase et al. 2017) [GMO16, CDG17]. The first practical implementations of MPC-in-the-head, achieving  $\approx 289$  KB and  $\approx 202$  KB for SHA-256, respectively. ZKB++ was used to build **Picnic**, a post-quantum digital signature scheme (NIST PQC Round 3 finalist), derived directly from MPC-in-the-head via Fiat–Shamir.

**Ligero++** (Bhadauria–Fang–Hazay–Venkatasubramaniam–Xie–Zhang 2020) [BFHVXY20]. Improved Ligero’s concrete efficiency and extended it to support more general constraint systems (R1CS).

**Aurora (Ben-Sasson–Chiesa–Riabzev–Spooner–Virza–Ward 2019) [BCR19].** Achieved  $\mathcal{O}(\log^2 s)$  proof size and  $\mathcal{O}(s \log s)$  prover time using FRI-based polynomial commitments — significantly smaller proofs than Ligerio at the cost of a more complex protocol.

**STARKs (Ben-Sasson–Bentov–Horesh–Riabzev 2018) [BBHR18].** Combined FRI with AIR constraints to achieve polylogarithmic proof size with no trusted setup, forming the foundation of Ethereum’s proof systems.

**VOLE-in-the-Head (Baum–Braun–Delpech de Saint Guilhem–Kloof–Orsini–Roy–Scholl 2023) [BBD23].** Replaced MPC commitments with VOLE correlations, giving publicly verifiable ZK with linear proof size and faster concrete verification than Ligerio and ZKB++.

**Impact on Post-Quantum Cryptography.** The MPC-in-the-head paradigm has become a primary approach to post-quantum ZK proofs, relying only on symmetric primitives (hash functions) immune to quantum attacks. Picnic [CDG17] reached NIST PQC Round 3, and the broader ecosystem continues to grow.

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